

Scalability & Future Plan

Date	11/07/2026
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	1 Mark

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No soil type input — model relies only on N, P, K, and climate values	Recommendations don't account for soil texture (sandy, loamy, clay, black), which affects real-world suitability	Medium

2	Free-tier hosting cold start	First request after inactivity takes 30-50 seconds while the Render server restarts	Low
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Scalability & Future Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs locally on the Flask development server.	Deploy on Render or another cloud platform with auto-scaling if user traffic increases.
2	Data Storage	Static CSV dataset and .pkl model files bundled directly in the repository	Move dataset and model artifacts to a managed database or object storage (e.g. PostgreSQL, AWS S3) if the dataset grows or needs frequent updates
3	Performance	Model loaded once at application startup; Flask handles prediction requests efficiently.	Deploy using Gunicorn with multiple workers and enable caching for improved concurrent performance.

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

3	Static dataset without real-time weather integration	Recommendations are based on historical averages, not real-time local weather conditions	Medium
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Scalability Plan:



4	Security	No authentication or rate limiting; relies on Render's default HTTPS	Add basic input sanitization, request rate limiting, and HTTPS enforcement if the app is opened to public/production traffic at scale
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Add soil type as an input feature (requires a richer dataset and model retraining), plus basic fertilizer recommendations	2-3 months	More personalized, soil-aware crop recommendations
Phase 3	Integrate a live weather API to auto-fill temperature, humidity, and rainfall based on user location	3-6 months	Recommendations reflect realtime conditions instead of historical averages
Phase 4	Add regional language support and a mobile-friendly progressive web app (PWA)	6-12 months	Wider accessibility for nonEnglishspeaking and mobile-first farmers